7-1 Computer Configuration

Point!

The Five Major Components of a Computer

(1) ([1]**Hardware**) : Devices such as the main computer unit and peripheral devices.

(2) Peripheral devices : Devices used by connecting them to a computer.
<Examples> Keyboard, mouse, display, printer, etc.

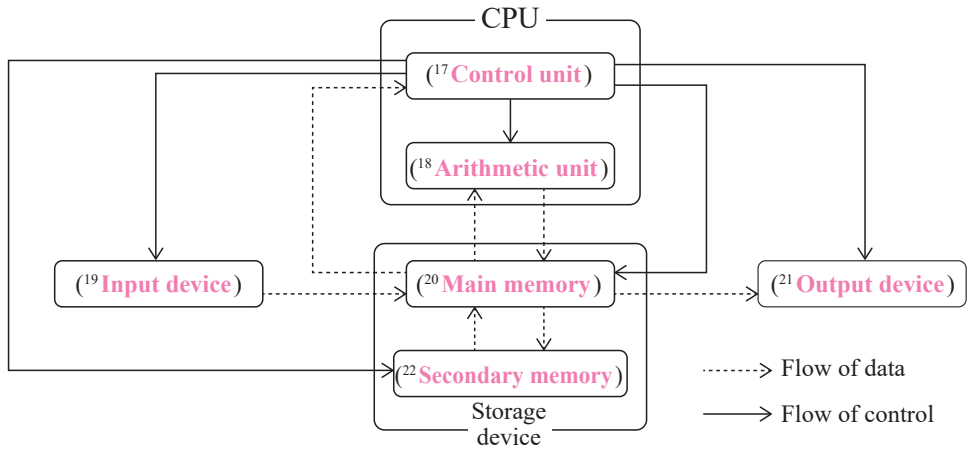
(3) The five major components of a computer : A computer is comprised of the following five components: ([2]**the control unit**), ([3]**the arithmetic unit**), ([4]**the memory unit**), ([5]**the input unit**), and ([6]**the output unit**).

- The control unit and arithmetic unit together are referred to as the ([7]**CPU (central processing unit)**).
- Storage devices are divided into ([8]**main memory**), which temporarily stores programs and data, and ([9]**secondary memory**), which is used for long-term storage.

<Roles of the Five Major Components of a Computer>

Five major components		Role	Example of major component
Control unit	CPU	Executes a computer's ([10]**instructions**) and issues ([11]**commands**) to each function.	CPU
Arithmetic unit		Performs ([12]**calculations**).	
Storage device	Main memory device (Memory)	([13]**Temporarily**) stores programs, data, etc.	Main memory
	Secondary memory (Storage)	For ([14]**long-term**) storage of programs and data.	Hard disk, SSD, USB memory
Input device		([15]**Inputs**) information from external sources.	Keyboard, mouse, scanner
Output device		([16]**Outputs**) information outside of the computer.	Display, printer

<Relationship of the Five Major Components of a Computer>



(4) ([23]**Interface**) : A component that mediates the exchange of information. It is used for connecting the main computer and peripheral devices.

[1] ([24]**USB**) : The most widely used interface for computer peripheral devices.
Compatible with printers, keyboards, and external hard drives.

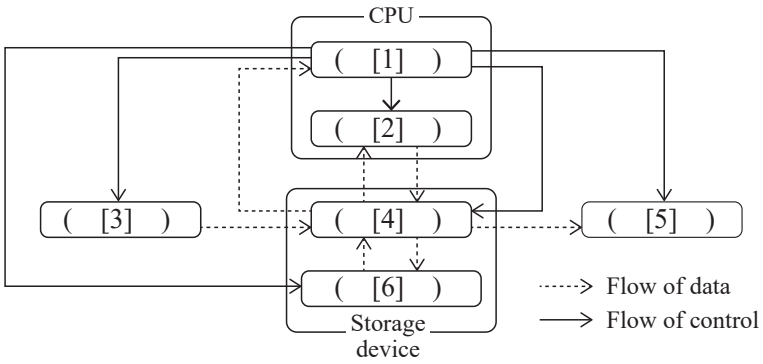
[2] ([25]**HDMI**) : A communication standard that allows video, audio, and other data to be transmitted through a single cable.
Compatible with digital televisions and audio equipment.

[3] ([26]**Ethernet**) : A communication standard used in wired LANs at home, the office, etc.
Compatible with devices such as hubs and routers.

Warm Up

Answer the following questions.

(1) The following diagram represents the relationship between computer components. Choose the appropriate device name that best fits into the blanks [1] to [6] from the options **A** to **F**, and answer using the letters.



A Input device **B** Output device **C** Control unit
D Arithmetic unit **E** Main memory device **F** Secondary memory

(2) Choose all of the following options from **A** to **H** that are output devices, and answer using the letters.

A Memory **B** Display **C** SSD **D** Mouse
E Keyboard **F** Printer **G** Hard disk **H** USB memory

Explanation

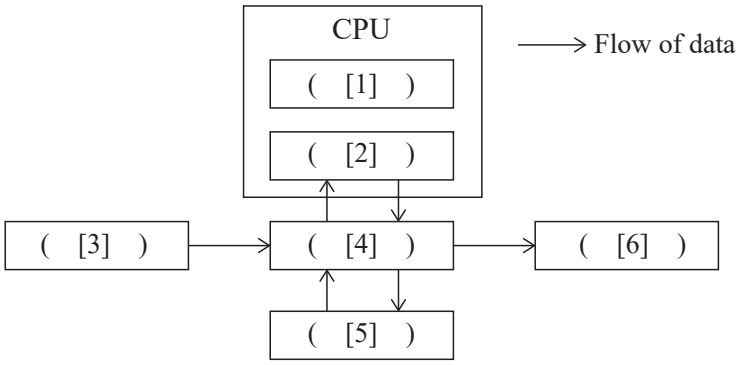
(1) [1] Control unit, because the flow of control is directed towards all devices. **C**
[2] Among the components of the CPU, [1] is the control unit; therefore, the arithmetic unit. **D**
[3] The starting point of the data flow; therefore, the input device. **A**
[4] Among storage devices, this is identified as the main memory because data flows directly into it from the input device. **E**
[5] Marks the end of the data flow; therefore, an output device. **B**
[6] Among storage devices, [4] is the main memory; therefore, the secondary memory. **F**

(2) **B, F**

Try

Answer the following questions.

(1) The following diagram represents the relationship between computer components. Choose the appropriate device name that best fits into the blanks [1] to [6] from the options **A** to **F**, and answer using the letters. Note that the order of blanks [1] and [2] does not matter.



A Input device **B** Output device **C** Control unit
D Arithmetic unit **E** Main memory device **F** Secondary memory

(2) Choose all computer components or peripheral devices corresponding to the following [1] to [5] from the options **A** to **I** below, and answer using the letters.

[1] Arithmetic unit, control unit [2] Main memory device [3] Secondary memory
[4] Input device [5] Output device

A Memory **B** Display **C** SSD
D Mouse **E** Keyboard **F** Printer
G Hard disk **H** USB memory **I** CPU

(3) Choose the term that best fits into the blanks [1] and [2] in the following sentences from the options **A** to **D** below, and answer using the letters.

([1]) is called the main memory and exchanges data directly with the CPU. ([2]) is also called secondary memory and serves the role of storing large amounts of data.

A Memory **B** Hardware
C Interface **D** Storage

(4) What is the interface described by each of the following [1] to [3]?

[1] An interface that connects devices to output video and audio through a single cable.
[2] The most commonly used interface for computer peripheral devices.
[3] Communication standard used for wired LANs in homes, offices, etc.

Exercise

1 Answer the following questions.

(1) Choose the most appropriate explanation of the control unit, which is one of the five major components of a computer from the options **A** to **D** below, and answer using the letter.

A It is a device that performs calculations.
B It is a device that executes a computer's instructions and issues commands to each function.
C It is a device for temporarily storing programs and data.
D It is a device for inputting information from external sources.

(2) Choose the names of the devices that correspond to computer devices and peripheral devices [1] to [5] from the options **A** to **E** below, and answer using the letters.

[1] Display [2] HDD [3] Keyboard
[4] CPU [5] Main memory

A Input device **B** Output device **C** Control unit, arithmetic unit
D Main memory device **E** Secondary memory

(3) Choose all the peripheral devices from the options **A** to **F**, and answer using the letters.

A Display **B** SSD **C** Memory
D CPU **E** Keyboard **F** USB memory

(4) Choose one correct statement regarding interfaces from the options **A** to **D** below, and answer using the letter.

A Devices that are not the computer itself but are used by connecting to a computer.
B Devices such as the computer itself and peripheral devices.
C Devices for temporarily storing programs and data.
D A component that mediates the exchange of information. It is used for connecting the main computer and peripheral devices.

(5) Choose the corresponding interface for each of the following peripheral devices [1] to [3] from the options **A** to **D**, and answer using the letters.

[1] Keyboard [2] Router [3] Audio equipment

A Ethernet **B** HDMI **C** SSD **D** USB

7-2 Computer Software

Point!

Hardware and Software

(1) Hardware : Devices such as the main computer unit and peripheral devices.
<Examples> CPU, memory, storage, etc.

(2) ([1]**Software**) : Programs and data that operate on hardware.

[1] ([2]**System software**) : Software necessary for operating the hardware.

[2] ([3]**Operating system (OS)**) : A type of system software responsible for the basic functions of a computer. An OS has management functions such as task management, memory management, and file management.
<Examples> Windows, macOS, Android OS, etc.

[3] ([4]**Application software**) : Software that operates on top of system software. Also known as application software.
<Examples> Word processing software, spreadsheet software, etc.

(3) ([5]**Device driver**) : A program that controls communication between a connected device and software on a computer. Also known simply as a "driver."

Warm Up

For the following statements **A** to **D** about an OS, mark "○" if the statement is appropriate, and "×" if it is inappropriate.

A Software used to perform specific tasks.
B Smartphones do not come equipped with an OS.
C A type of system software essential for operating the hardware.
D Application software cannot run without system software.

Explanation

A: An explanation about application software. Therefore, ×
B: Smartphones are equipped with an OS. For example, iPhones are equipped with iOS, while Android phones run on Android OS. Therefore, ×
C: An OS is a type of system software responsible for the basic functions of a computer. Therefore, ○
D: Application software is software that operates on top of system software. Therefore, ○

Try

Answer the following questions.

(1) Complete the following sentences by filling in the blanks [1] to [3] with the appropriate terms.

> Types of software include ([1]) software and ([2]) software. ([1]) software includes ([3]). In the case of computers, examples are Windows and macOS. Additionally, ([2]) software includes word processing software and spreadsheet software.

(2) What is the term for software that is used to control and operate devices connected to a computer?

(3) For the following statements **A** to **D**, answer with "a" if the statement is about operating systems or "b" if the statement is about application software.

A A program executed on a computer that is designed for a specific purpose.
B A program that controls software and hardware on a computer system.
C Includes word processing software and spreadsheet software.
D Included in the system software.

Exercise

1 Answer the following questions.

(1) Choose all the devices and software corresponding to the following [1] to [3] from the options **A** to **F** below, and answer using the letters.

[1] Operating system [2] Hardware [3] Application software

A Memory **B** Spreadsheet software **C** Windows
D macOS **E** CPU **F** Document processing software

(2) Choose one correct statement regarding device drivers from the options **A** to **D** below, and answer using the letter.

A A type of system software. Examples include, Windows, macOS, etc.
B Also referred to as application software.
C A program that controls communication between the devices connected to a computer and the software.
D The software necessary for operating hardware.

7-3 Logic Circuits

Point!

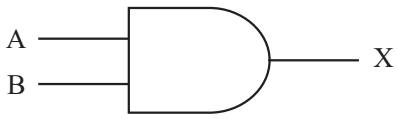
1 Logic Circuits

(1) ([1]**Logical operations**) : Operations performed using combinations of the numbers 0 and 1. In computers, "1" is processed as true and "0" as false.

(2) ([2]**Logic circuit**) : A circuit designed to perform logical operations.

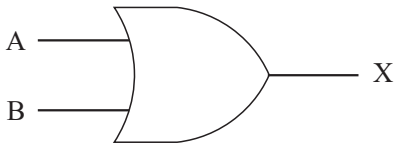
(3) ([3]**Truth table**) : A table that shows all possible combinations of inputs and outputs for a logic circuit.

(4) ([4]**AND gate (logical conjunction circuit)**) : A circuit that outputs 1 only when all inputs are 1.



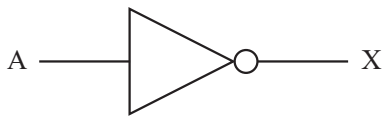
Input		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

(5) ([5]**OR gate (logical disjunction circuit)**) : A circuit that outputs 1 if at least one of the inputs is 1.



Input		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

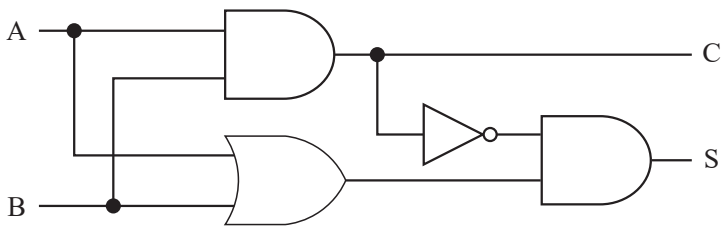
(6) ([6]**NOT circuit (inverter circuit)**) : A circuit that outputs the opposite result of the input.



Input	Output
A	X
0	1
1	0

2 Half Adder Circuit and Full Adder Circuit

(1) ([7]**Half adder circuit**) : A circuit representing the addition of single-digit numbers, composed of AND, OR, and NOT gates.



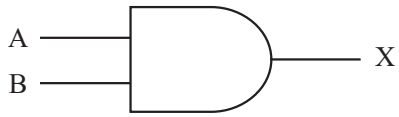
Input		Output	
A	B	C	S
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	0

(2) ([8]**Full adder circuit**) : A circuit that considers carry from the lower bit and carry to the higher bit.

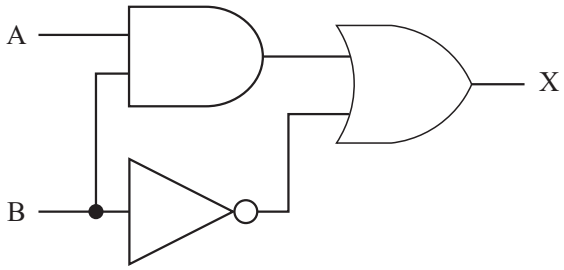
Warm Up

Answer the following questions.

(1) Name the following logic circuit and create a truth table.



(2) A circuit was created by combining logic circuits as shown in the following diagram. In this case, create a truth table for inputs A, B, and output X.



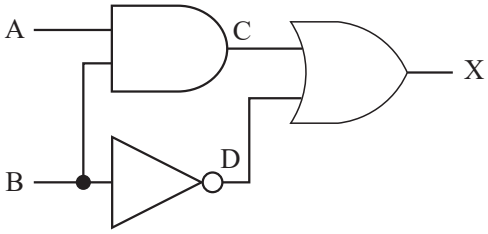
Explanation

(1) AND Circuit (logical conjunction circuit).

An AND circuit is a circuit that outputs 1 only when all inputs are 1. Therefore, the truth table is as follows.

Input		Output
A	B	X
0	0	0
0	1	0
1	0	0
1	1	1

(2) Create and analyze the truth table for each circuit.



A	B	C	D	X
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

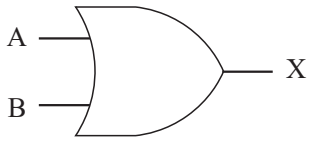
Therefore, the truth table is as follows.

Input		Output
A	B	X
0	0	1
0	1	0
1	0	1
1	1	1

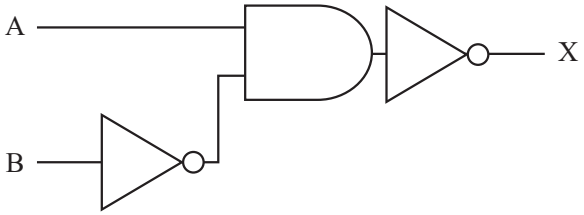
Try

Answer the following questions.

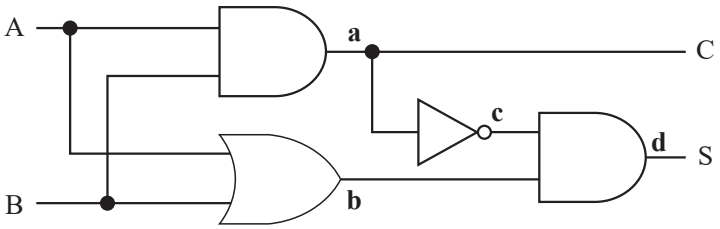
(1) Name the following logic circuit and create a truth table.



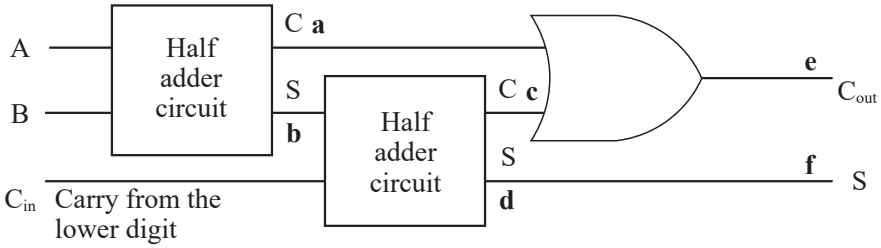
(2) A circuit was created by combining logic circuits as shown in the following diagram. In this case, create a truth table for inputs A, B, and output X.



(3) In the following half adder circuit, determine the values of each output **a** to **d** when input A is 0 and input B is 1.

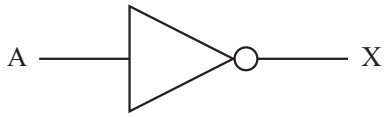


(4) In the following full adder circuit, determine the values of each output **a** to **f** when input A is 1, input B is 1, and input C_{in} is 0.

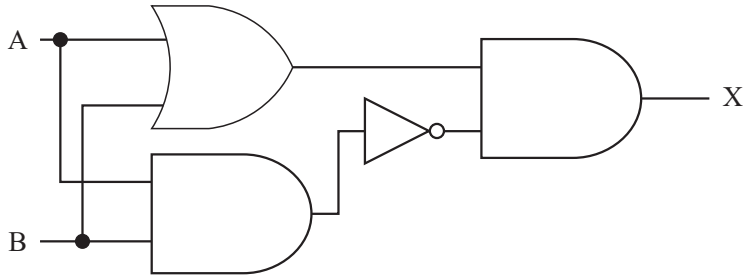


1 Answer the following questions.

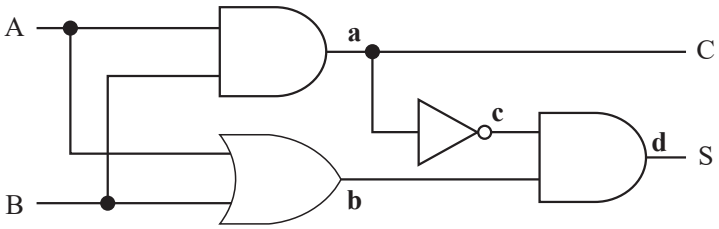
(1) Name the following logic circuit and create a truth table.



(2) A circuit was created by combining logic circuits as shown in the following diagram. In this case, create a truth table for inputs A, B, and output X.



(3) In the following half adder circuit, determine the values of each output **a** to **d** when input A is 1 and input B is 1.



(4) In the following full adder circuit, determine the values of each output **a** to **f** when input A is 1, input B is 0, and input C_{in} is 1.

